Wave-Solder — Troubleshooting Guide (Faultiest Station on Line-E)

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> **Audience:** Maintenance Tech (primary) · Line Manager · Quality Worker **Applies to:** Line-E, station 6, Wave-Solder

1. Purpose

Wave-Solder bonds the through-hole leads inserted at station 5. With a 0.7 % per-cycle fault probability — the **highest on Line-E** — it is the single most common cause of downtime on the electronics line. This troubleshooting guide is the fast path from a `Fault` telemetry event back to `Running` for the three tracked fault types.

2. Normal operating envelope

- Ideal cycle time: **35 s**
- Max acceptable cycle time: **44 s**
- Expected reject rate: **≤ 3 %**
- Fault probability per cycle: **0.7 %** (line max)
- Input buffer capacity: **5 parts**

3. Idle reasons

`idle_reason`	Meaning here	First check
`Starved`	Through-Hole-Insert (station 5) is not feeding boards.	Verify Through-Hole-Insert state.
`Blocked`	Functional-Test (station 7) buffer full.	Verify Functional-Test state.

Because of this station's high fault rate, **most downstream `Starved` events on Line-E trace back to this station**. When triaging starvation on stations 7 or 8, check Wave-Solder first.

4. Faults & corrective actions

4.1 `Solder-Temp-Drift`

- **Symptoms in telemetry:** `fault_type = "Solder-Temp-Drift"`. Joint quality rejects rise at Functional-Test; cold-joint codes cluster.
- **Likely root cause:** thermocouple drift, heater control failing.
- **Corrective action:**
  1. Verify pot temperature against a calibrated probe.
  2. If drift > 5 °C, calibrate or replace the thermocouple.
  3. Inspect heater elements; replace if degraded.
- **Restart criteria:** pot temperature stable within ± 3 °C of setpoint for 10 minutes.

4.2 `Flux-Low`

- **Symptoms in telemetry:** `fault_type = "Flux-Low"`. Wetting defects rise; some leads show open joints.
- **Likely root cause:** flux reservoir near-empty; spray nozzles clogged.
- **Corrective action:**
  1. Refill flux reservoir.
  2. Inspect and clean spray nozzles.
  3. Verify flux coverage on a test board.
- **Restart criteria:** uniform flux coverage observed on the test board.

4.3 `Conveyor-Speed-Fault`

- **Symptoms in telemetry:** `fault_type = "Conveyor-Speed-Fault"`. Time in the wave is wrong; large solder defect spike.
- **Likely root cause:** drive motor encoder failure, or VFD tripped.
- **Corrective action:**
  1. LOTO. Inspect encoder cable and connection.
  2. Verify VFD fault code; reset or replace.
- **Restart criteria:** conveyor speed stable across 5 minutes.

5. Preventive maintenance schedule

- **Daily:** dross removal; flux check; profile log.
- **Weekly:** clean spray nozzles; inspect fingers.
- **Monthly:** solder analysis for contamination.
- **Quarterly:** full conveyor service.

6. References

- `Maintenance_Workflow_Overview.pdf`
- `Safety_Lockout_Tagout_Procedure.pdf`
- `OEE_Targets_and_Escalation.pdf`
- `Line-E_05_Through-Hole-Insert_SOP.pdf`
- `Line-E_07_Functional-Test_SOP.pdf`